Fabrication and Integration Process

1. **Material Choice Justification :**

For the Snake Path:

1. Snake Path Walls — Mountboard

The maze walls stand at a significant height relative to their thickness. Mountboard has a dense, compressed cardboard core that gives it excellent **column strength,** it resists buckling and toppling under its own weight without needing additional support. A thinner material like chart paper or foam sheet would collapse or bow outward at this scale.

Our robot uses ultrasonic sensors for wall detection, mountboard's **smooth, non-porous surface** provides consistent reflection. Corrugated or textured surfaces scatter signals unpredictably.

Compared to MDF or acrylic, mountboard is extremely **lightweight**, so the walls don't add significant load to the base, preventing warping or joint failure over the 1.2m × 1.8m footprint.

Why not MDF? *Too heavy, at this scale, MDF walls would require mechanical fasteners and significantly stress the base.* ***Why not acrylic?*** *Cost-prohibitive for the sheer linear length of walls needed, and acrylic is brittle at thin sections.*

2. Base — Sunboard

At 1.2m × 1.8m, the base needs to remain **perfectly flat** across a large unsupported area. Sunboard (foam board) has a polystyrene core bonded between two smooth paper faces, which resists sagging over span far better than cardboard or chart paper at this size.

The robot's wheels make direct contact with the base. Sunboard's **smooth paper face** offers a uniform, low-friction surface, ensuring predictable wheel traction and consistent motor performance, avoiding the speed variations we faced on a rough or uneven surface.

A 1.2m × 1.8m base in MDF would weigh upward of 8–10 kg, making it nearly impossible to transport. Sunboard keeps the **entire snake path portable and handleable** by one person alone. (Evidence inserted)

Why not corrugated cardboard? *Compresses under load and buckles over large spans, creating an uneven rolling surface.*

4. Robot Body & Top Cover — Chart Paper & Mountboard

Every gram added to the robot body is a gram the motors must work harder to move. Chart paper and mountboard are among the **lightest sheet materials available**, keeping the robot's total mass low and extending effective battery life per run.

During prototyping and testing, our sensor positions and component layouts change frequently. Chart paper and mountboard can be **cut, re-cut, and replaced in minutes** with scissors. Acrylic or MDF would require re-cutting the entire panel.

Mountboard panels are stiff enough to hold their shape as a top cover without sagging onto the circuitry below, while remaining far lighter and cheaper than acrylic for the same panel area.

Why not full acrylic body? *Unnecessary cost and weight for parts that bear no structural function.* ***Why not corrugated cardboard?*** *Too bulky and dimensionally inconsistent for a compact robot form factor.*

2. Fabrication Steps :

Maze Construction

Step 1 — Material Procurement & Sizing Sourced 2 full sheet of sunboard joined them (1.2m × 1.8m) for the base and multiple mountboard sheets for wall construction. Verified flatness of the sunboard sheet by laying it on a level surface before any cutting began.

Step 2 — Base Preparation The sunboard sheet was kept at its full 1.2m × 1.8m dimension as the maze base. Edges were lightly trimmed with a steel ruler and craft knife to ensure all four sides were perfectly square and clean.

Step 3 — Snake Path Layout Drafting The snake path and wall layout were drawn to scale on paper first, then wall positions were lightly pencil-marked directly onto the sunboard base using a ruler and set square. This ensured corridor widths were consistent and wide enough to accommodate the robot's chassis with sufficient navigational clearance, and enough space for it to turn around at dead ends.

Step 4 — Mountboard Wall Cutting Mountboard sheets were measured and marked according to the wall segment lengths from the layout plan. Each wall segment was cut using a metal ruler and sharp craft knife on a cutting mat, applying multiple light passes to maintain clean, perpendicular edges without tearing the board face.

Step 5 — Wall Dry Fitting All cut wall segments were suck onto the marked base with adhesive to verify alignment, corridor widths, and path continuity.

Step 6 — Wall Joining & Assembly Each wall segment was fixed upright onto the sunboard base using fevibond. Glue was applied to the bottom edge of each mountboard wall and pressed firmly onto the base, allowing it to penetrate slightly into the foam core for a strong bond. Each segment was held vertically for approximately 30

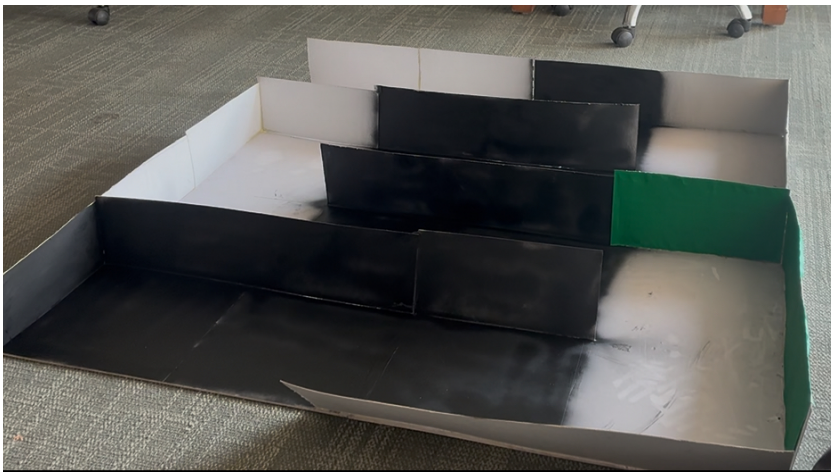
seconds until set.

Step 7 — Spray Painting the Board Black Once the entire snake path structure was assembled and all joints were fully cured, the whole snake path walls and base, was spray painted black. This was done in a ventilated area, with even sweeping strokes held at a consistent distance from the surface to achieve a uniform coat across all wall faces and the base floor. Multiple light coats were applied rather than one heavy coat to prevent dripping and surface bubbling on the sunboard. The maze was left to dry completely between coats.

Step 8 — Applying Coloured Chart Paper After the spray paint had fully dried, selected sections of the maze were decorated by cutting plain chart paper into the required shapes and sizes and adhering them to the designated wall faces and floor areas using glue. The chart paper was pressed down firmly and smoothed out to

eliminate air bubbles and ensure full surface contact, giving the maze its final coloured visual appearance against the black painted background.

Step 9 — Final Maze Inspection The completed maze was checked for wall verticality using a small set square at multiple points. The floor was inspected for any raised chart paper edges or glue residue that could obstruct wheel movement, and any such areas were pressed flat or trimmed.



Robot Chassis Assembly

Step 1 — Procurement of Pre-Made Chassis The robot chassis, including the base plate, motor mounts, motors, wheels, and axle hardware, was procured as a ready-made kit.

Step 2 — Chassis Assembly & Motor Fitting The chassis components were assembled according to the kit structure. The two drive motors were mounted into their designated brackets on the chassis base and secured with the provided bolts and nuts. Care was taken to ensure both motors were seated symmetrically so the wheel track was even on both sides.

Step 3 — Wheel & Axle Fitting Wheels were fitted onto the motor output shafts and secured. The assembled chassis was placed on a flat surface to confirm both wheels made even ground contact, verifying the axles were parallel and the robot sat level without rocking.

Step 4 — Electronics Mounting The microcontroller board, motor driver module, and sensor array were positioned on the chassis and secured using tape where available, or double-sided foam tape.

Step 5 — Wiring & Cable Management All wiring between the battery, motor driver, microcontroller, and sensors was connected and routed tidily. Excess wire length was folded and secured with tape to prevent loose wires from catching on snake path or interfering with wheel rotation.

Step 6 — Battery Placement The battery pack was placed at the longitudinal centre of the chassis to balance weight distribution between front and rear, reducing the risk of tipping during sharp turns and acceleration within the maze corridors.

Robot Body & Cover Construction

Step 1 — Body Panel Templating With the chassis fully assembled, the dimensions of each side of the robot were measured. Templates for the four side body panels were drawn out on chart paper, accounting for the chassis height and perimeter length on each face.

Step 2 — Side Panel Cutting The chart paper and mount board panels were cut to the templated dimensions using scissors and a craft knife. Straight edges were ensured by cutting against a ruler so panels would sit flush and even around the chassis.

Step 3 — Top Cover Cutting The top cover was measured and cut from chart paper, sized to span the full top opening of the body with a small overlap on each edge to sit neatly on the side panels.

Step 4 — Body Assembly & Attachment The side panels were attached around the chassis using hot glue applied sparingly along the edges. Each panel was held in position until the glue set. Alignment was checked after each panel to ensure no face was skewed outward, which could cause the robot to catch on maze walls during navigation.

Step 5 — Top Cover Attachment The chart paper top cover was glued onto the top edges of the side panels, completing the enclosure around the electronics. A small lift tab was incorporated at one end to allow quick access to the battery and wiring without needing to dismantle the body panels.